

AY 2025-26

Data Analytics with R

Quality Laboratory Manual

Experiment No. 01

To Explore and classify various types of datasets with their characteristics and applications in different domains



**Course Instructor –
Dr. Tahseen A. Mulla**

Experiment No. 01

Title of Experiment: To explore and classify various types of datasets with their characteristics and applications in different domains

Aim of Experiment: To understand, explore and classify different types of datasets based on their structure, nature, source and domain and to identify their characteristics and real-world applications using basic data exploration techniques

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Internet Connection (for dataset access)

Experiment Objectives:

- To understand different types of datasets used in data analytics
- To classify datasets based on structure, data type and domain
- To identify key characteristics of each dataset type
- To explore sample datasets using R
- To map datasets to real-world applications across domains

Experiment Outcomes:

After completing the experiment, students will be able to –

- Distinguish between structured, semi-structured and unstructured data
- Identify numerical, categorical, time-series and text datasets
- Relate dataset types to applications in healthcare, finance, education and social media
- Develop analytical thinking required for further data analytics experiments

Theory:

Data is the heart and soul of business analytics. It can be defined as information that is collected, processed, and analysed to obtain data insights and make informed decisions. This data can be collected from different sources, including social media platforms, web sources like websites and webpages, financial transactions, and other channels running on the internet.

Types of Data

Some common categories of Data are as follows –

1. Structured Data

Structured data is a type of data that is highly organized and easily searchable using simple algorithms; it is easily stored and managed using traditional data management tools like spreadsheets, SQL databases, or tables. In other words, we can say that the structured data is organized into rows and columns and then can easily be searchable as and when required. Structured data is often quantitative and numeric which may include numbers, percentages, and related data. Structured data is comparatively simple to analyze using statistical techniques such as regression analysis, correlation analysis etc.

Examples of Structured Data

Some of the most relevant examples of Structured Data are as follows –

- Spreadsheets – Data organized in Excel sheets in tabular form or the form of rows and columns.
- Google Sheets – Google Sheets are organized data on the cloud.
- Relational Databases – As its name suggests relational databases store data in tabular form. Some common examples of relational databases are MySQL, PostgreSQL, and Oracle.
- Data Warehouses – Large-scale storage of structured data for analysis and reporting purposes, such as Amazon Redshift or Google BigQuery.
- CSV (Comma-separated values) Files – In CSV files, each line represents a row and each field within the line is separated by a comma.

Characteristics of Structured Data

Some of the common characteristics of Structured Data are as follows –

- Follows a predefined Schema – It follows a predefined schema or predesigned format that defines the data types, relationships, and limitations
- Organized in Tabular Form – Structured data is typically arranged in rows and columns. For example, a spreadsheet, table or database. The data is well organized using data definition, format, and meaning of the data. The data lives in fixed fields inside a record or a file
- Consistent Data Types – In a tabular form of data, each column has a structured data table which includes specific data types, such as integer, string, or date
- Easily Searchable – Structured data may be efficiently searched, queried, and modified using SQL
- Grouped – Similar entities are grouped to form relations or classes. The data is simple to access and query. As a result, other programs can easily use it

Advantages of Structured Data

Some of the key advantages of Structured Data are as follows –

- Easy to store and access Structured data has a predefined structure which makes it easy to understand, store and access data
- Efficiency Structured data can easily be stored, retrieved, processed, and managed using traditional and advanced database management systems efficiently
- Accuracy Data constraints and validation are imposed on structured data to maintain its integrity and accuracy
- Scalability Well-suited for large-scale data storage and complex queries
- Interoperability It can be integrated and used with a variety of business intelligence and

- reporting solutions
- Requires less storage space: It requires less storage space to store data

Disadvantages of structured data

- A predefined structure of data is limited to use
- It has limited storage functions
- Difficult to change or update; this results in massive expenditure of resources and time

2. Unstructured Data

Unstructured data is data which doesn't contain structure, predefined format or schema to store the data. Unstructured data is very challenging to store, and process using traditional relational databases or RDBMS. Unstructured data includes business documents, email messages, videos, images, webpages, and audio files

It is often qualitative, i.e. descriptive and narrative. Customer credit reports, insurance claims, and airline ticket complaints are some of the key examples of unstructured textual data that have commercial implications. Unstructured data can be analyzed using advanced analytics techniques such as natural language processing (NLP) for sentiment analysis

Examples of Unstructured Data

Unstructured data is qualitative rather than quantitative, which means that it is more based on its characteristics and categorical.

- Emails
- Social media posts
- Audio and video files
- Sensor data
- Memos
- Documents (PDFs, Word files)
- Webpages
- Images (JPEG, GIF, PNG, etc.)

Characteristics of Unstructured Data

Some of the key characteristics of unstructured data are as follows –

- No specific Data model – Unstructured data doesn't have any specific data model; which means it doesn't have a specific format and structure to store data
- Volume – Volume refers to the size of data; modern datasets come in bigger sizes which means they have a large volume of data
- Variety – Unstructured data includes different forms of data like text, multimedia, etc.
- Doesn't have semantics – Unstructured data doesn't have specific rules and regulations
- Complexity – Hard to manage and analyses with traditional data tools
- Storage – Typically stored in data lakes or NoSQL databases

Advantages of Unstructured Data

Some of the key advantages of Unstructured Data are as follows –

- **Rich Source of Information** – Unstructured data is a rich source of information. It contains in-depth information, capturing nuances and context that structured data misses.
- **Variety of information** – Unstructured data contains a variety of information.
- **Provides comprehensive insights** – Unstructured data provides comprehensive insights into customer sentiments, behaviors, and preferences
- **Flexible Diverse Sources** – The flexibility of unstructured data allows it to include a wide range of data formats, such as text, images, and videos
- **More Detailed Information** – Unstructured data can contain more precise and granular information, including nuances, feelings, and specific details that may be lost in structured data
- **Real-time Data** – Can be generated and analyzed in real-time
- **Deeper Analysis using AI/ML** – AI/ML is used to analyse unstructured data

Disadvantages of Unstructured Data

Some of the key disadvantages of Unstructured Data are as follows –

- **No standard structure** – Unstructured Data doesn't have a predefined structure to store, process and access the data
- **Inconsistent in format and content** – Data from different sources may be inconsistent in format and content, complicating analysis efforts
- **Complexity in Analysis** – due to the lack of structure; it has complexity to analyse data. It uses complex algorithms to process the data
- **Performance Issues** – Querying and retrieving specific information can be slower
- **Noise and irrelevant information** – It may contain noise and irrelevant information which may increase challenges to ensure data quality and consistency

3. Semi-Structured Data

Semi-structured data combines feature of both structured and unstructured data. This sort of data includes information that is partially ordered but not enough to be categorized as structured data. Semi-structured data includes XML and JSON files, which are organized and also contain unstructured data elements. Semi-structured data is often analyzed using a combination of traditional data management tools and sophisticated analytics techniques

Semi-structured data applies to a variety of applications where some degree of organization is desirable but stringent schema requirements are not necessary. Hence, it falls between structured and unstructured data

Examples of Semi-Structured Data

Some common examples of Semi-Structured Data are as follows –

- XML (eXtensible Markup Language) Files
- JSON (JavaScript Object Notation) Files
- Email
- HTML (Hypertext Markup Language) Documents
- Log Files
- NoSQL Databases
- Sensor Data

Characteristics of Semi-structured Data

Some common characteristics of Semi-structured Data are as follows –

- Partially structured – Semi-structured data is partially structured; it means it's a combination of structured and unstructured data.
- Flexible Schema – Semi-structured data doesn't have any specific structure. Hence, it doesn't conform to a data model.
- Self-describing Nature – Data often contains metadata or tags that describe its structure and significance. XML and JSON are some examples.
- Easier Data Integration – The flexible schema makes it easy to combine semi-structured data from different sources.
- Supports Complex Data Types – It supports complex data types like arrays and objects.

Advantages of Semi-structured Data

Some of the common advantages of Semi-structured Data are as follows –

- Flexibility – Semi-structured data may contain different data types and formats.
- Flexible schema for Data Integration – The flexible schema of semi-structured data allows its users to integrate data collected from different sources.
- Scalability – Semi-structured data has features to store data in a scalable manner.
- Interoperability – It includes files like JSON, XML, and YAML.
- Complex Data Types – Semi-structured data can handle arrays, objects, and other complex data types, enabling the representation of rich, multi-dimensional data.
- Storage Efficient – Semi-structured data can be more storage-efficient.

Disadvantages of Semi-structured data

Some of the common disadvantages of Semi-structured Data are as follows –

- Partial structured – It contains partial structure data which can be a bit difficult to store and process
- Inconsistencies in the data – The lack of a strict schema can result in inconsistencies in the data
- Complexity in Data Management – Managing semi-structured data can be complex due to the lack of a fixed schema
- Performance Issues – Querying and processing semi-structured data may be less efficient than structured data
- Limited Tool Support – There are limited tools available for managing and analysing semi-structured data

Task to Perform

Case Study-Based Observation Task: Dataset Exploration and Classification for a Smart City Analytics Project

A municipal corporation of a smart city plans to improve urban living by using data analytics. The city collects data from multiple sources such as traffic sensors, hospitals, educational institutions, social media platforms and weather stations. The collected data will be analyzed to support decision-making in areas like traffic management, public health monitoring, academic performance improvement and citizen sentiment analysis

As a data analytics trainee, you are assigned the task of exploring and classifying the datasets used in this smart city project before actual data analysis begins

Datasets provided –

- 1) Bangalore_traffic_Dataset.csv
- 2) Hospital_Patient_Records.csv
- 3) Students_Performance.csv
- 4) Telangana_Weather_Data.csv
- 5) Noise_Complaints.csv

Student Task to be Submitted –

Students are required to perform the following tasks –

1) Dataset Classification

Classify each dataset based on structure (Structured/ Semi-structured / Unstructured), its data type (Numerical/ Categorical/ Time-Series/ Text), source (Sensor-generated/ Institutional/ Social Media)

2) Dataset Characteristics

Identify key characteristics such as size and frequency of data, nature of variables, presence of time dependency, possible data quality issues

3) Domain Mapping

Identify the application domain for each dataset for traffic management, healthcare, education, weather forecasting, public opinion analysis

4) Analytics Perspective

Suggest one possible analytics task for each dataset (Eg: prediction, classification, trend analysis, sentiment analysis)

Expected Observation Table Format –

Dataset Name	Structure	Data Type	Domain	Key Characteristics	Possible Analytics Task
Bangalore_traffic_Dataset.csv					
Hospital_Patient_Records.csv					
Students_Performance.csv					
Telangana_Weather_Data.csv					
Noise_Complaints.csv					

Submission Guidelines –

- 1) Observation report should be at least 2-3 pages
- 2) It should include –
 - a. Dataset classification table
 - b. Written explanations for each dataset
- 3) Submit the experiment in PDF format

Observations:

- Structured datasets are easier to analyze using data frames in R
- Numerical datasets allow statistical analysis such as mean, median and variance
- Categorical domains demand different preprocessing techniques
- Different domains demand different preprocessing techniques
- Dataset characteristics directly influence the choice of analytics techniques

Conclusion:

In this experiment, various types of datasets were explored and classified based on structure, data type, source and application domain. The experiment helped in understanding how dataset characteristics influence analytical approaches. This foundational knowledge is essential for performing effective data analysis and building predictive models using R.

Expected Oral Questions:

- 1) What is a dataset?
- 2) Differentiate between structured and unstructured data
- 3) Give examples of numerical and categorical data
- 4) What is time-series data?
- 5) Why is dataset classification important in data analytics?
- 6) Name some open-source dataset repositories
- 7) How does domain knowledge help in data analysis?

FAQs in Interview:

- 1) Why is understanding dataset types important before analysis?
- 2) Can one dataset belong to multiple categories?
- 3) What type of data is social media data?
- 4) Which R function is commonly used to explore datasets?
- 5) Which domain uses time-series data extensively?